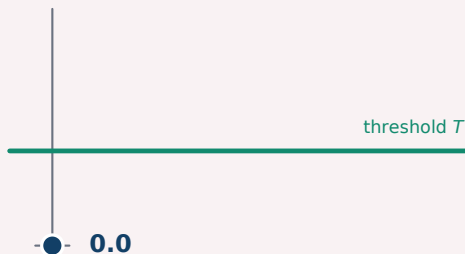


A non-detection is an upper bound, not a zero

The same measurement, the same legal threshold, read two ways

What the record stores

a number, indistinguishable from a measured zero

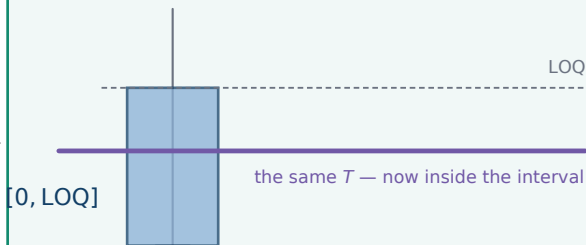


Compliant

one of two possible verdicts

What the measurement established

an interval: the analyte may be present below the limit



Cannot be determined

a verdict the two-valued model cannot express

CENSO binds the quantification limit to the analytical run rather than to the instrument, so the interval survives into the data.

45%

of 33 million European samples
lie below the quantification limit

25%

of station-years record neither a
censoring flag nor a limit

40%

of assessments against a European
standard use a method failing the
legal LOQ criterion (2009/90/EC)

EEA Waterbase: 4.2 million river station-years, 637 substances, 37 countries.

The share recording no limit ranges from 3 % to 97 % between countries — reporting practice, not chemistry.